Coble Paper Outline

0: Introduction

- **Main Theorem A:** For F_{Co}, the (normalization of the) KSBA compactification is semitoroidal for an explicit coarsening/refinement of the Coxeter chamber. (E.g. union of 7 chambers, subdivision of fundamental chamber into 4 subchambers, etc.)

- **Main Theorem B**: Cusp correspondence involving $F_{Co}\to F_{En}$ and $F_{Co, 2} \to F_{En, 2}$; this defines $F_{Co, 2}$, which is not in the current literature.

- **Main Theorem C:** How Coble diagrams are obtained from Enriques diagrams; something like folding K3 diagrams.

- **Main Theorem D:** How the IAS for Cobles are obtained from the IAS for Enriques.

- **Main Theorem E:** The KSBA stable degenerations at each 0-cusp of F_{Co} admit explicit descriptions: the irreducible components are (potentially) ABCDEFGH surfaces which are determined by the IAS in Theorem D.

1: Introduction

- *Historical motivation*: why Coble looked at these, what he proved. Classification in DZ99. Dimension of moduli space. Appearance as the boundary of F_{En}. Other constructions of moduli spaces (e.g. GIT). Discussion of n boundary components and relationship to K3 moduli involving Halphen pencils. Importance of Cremona special point sets in terms of size of automorphism groups. Direct relationship to Enriques by allowing the K3 cover to acquire an ODP in a degeneration.

- *Definition*: Coble surfaces of K3 type.

- *Proposition*: the KSBA compactification exists and satisfies standard (?) nice properties

2: Geometric Construction

- *Remark*: adapting the Horikawa branched cover of $P^1\times P^1$ construction to attain Cobles. The main square diagram. Description of the ramification divisor of the K3 cover.

- *Remark*: Using the ramficiation divisor to construct a KSBA moduli space. Direct link to nonsymplectic involution moduli spaces.

- *Remark*: relevant lattices and period domain constructions of F_{En} and F_{Co}. Discuss relation of F_{Co} to H_{-2} in F_{En}. Remark on nodal Enriques surfaces as H_{-4}.

2: F_{En} theory from AE, AET, AEH, AEGS, but only the KSBA aspects

- *Definition*: K3 surfaces and h.e. K3s

- *Construction*: of $F_{(2,2,0)}$ that uses new symplectic involution machinery

- *Definition*: semitoroidal compactifications

- **Theorem (Cite)**: for $F_{(2,2,0)}$, KSBA = semitoroidal with explicit fans

- **Theorem (Cite):** the maximal KSBA stable degeneration for F_{En} admits an explicit description: the irreducible components are ADE surfaces which are encoded by subdiagrams of Coxeter diagrams at the cusps.

- 3: Semitoroidal compactifications,

  - *Remark*: basic BB theory, correspondence between isotropic flags and boundary components, why we consider BB when we want KSBA. Looijenga's semitoroidal generalizations.

  - *Definition*: cusp diagram, how to read the geometry off of it.

  - *Proposition*. BB for F_{En} and its cusp diagram. Explicit identification of isotropic vectors and planes.

  - *Proposition*. BB for F_{Co} and cusp its diagram. Explicit identification of isotropic vectors and planes.

  - **Main Theorem B**: Cusp correspondence $F_{Co}\to F_{En}$.

    - *Proof sketch*: Explicit calculation of divisibilities. Cite theorem that divisibilities determine the correspondences.
    - Remark on matching modular curves.
    - Remark on matching maximal parabolic subdiagrams.

- 4: Coxeter diagrams

  - *Definition*: Coxeter diagrams, node/edge conventions and how to read the diagram

  - **Main Theorem C:** Coble diagrams are obtained by some combinatorial modification of Enriques or K3 diagrams, which has geometric significance.

- 5: IAS for the single Coble cusp

  - **Main Theorem D:** IAS for F_{Co} are obtained by specializing IAS for F_{En} in some way.

  - **Lemma**: a degeneration in F_{Co} of disc type is described by an integral affine structure on $\mathbb{D}$ with charge 12.

  - *Construction*: Describe the parameter choices ℓ_i giving 9-dim families of IAS^2s

  - *Construction*: Describe the Symington polytopes P (with nontoric blowups) and verify the IAS^2s have charge 12.

  - *Examples*: include 2 explicit examples of IAS for Cobles.

- 6: KSBA Stable Models for E_2

  - **Main Theorem E:** The maximal KSBA stable degenerations at each 0-cusp of F_{Co} admit an explicit description: the irreducible components are ABCDE surfaces determined by the IASs in **Theorem D**. Which ones are pumpkin vs smashed pumpkin type. Enumerated tables of all ADE surface possibilities.

    - *Proof*: Analysis of how hemispherical IAS singularities collide with equatorial singularities under a disc-type quotient of the sphere.

  - **Main Theorem A:** For E_2, KSBA = semitoroidal with explicit fans/chambers (e.g. a subdivision of the Coxeter chambers into N subchambers)

    - *Proof sketch*: fan vs semifan follows from a finiteness condition on a subdiagram of the Coxeter diagram. What the actual fan/semifan is: an analysis that I do not know how to carry out yet.